

ORÓMA: An Offline-First Persistent Episodic Memory Architecture for Edge Cognitive Agents

Jörg Werner

Independent Researcher, Germany

ORÓMA Project

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Abstract

ORÓMA (Offline-Realtime-Organic-Memory-AI) is an offline-first adaptive edge intelligence architecture designed for persistent operation on resource-constrained hardware. In contrast to cloud-centered, scale-driven AI systems, ORÓMA emphasizes local autonomy, persistent episodic memory, replay-based consolidation, and continual adaptation under real-world runtime constraints.

The architecture is centered around Snap- and SnapChain-based episodic structures, a Day/Dream execution model, replay-driven consolidation, and binding-oriented mechanisms for relating events, contexts, and multimodal signals across time. ORÓMA is implemented as a long-running edge system and has been exercised across multiple domains including game-like environments, replay workflows, and embodied perception/control scenarios such as PTZ attention.

This document presents the conceptual structure, core mechanisms, and current implementation direction of ORÓMA as a persistent edge cognitive architecture. The purpose of this publication is to document the architectural novelty, terminology, and system principles as prior art and as a foundation for future technical and scientific work.

Keywords: edge AI, offline AI, continual learning, episodic memory, replay learning, adaptive memory systems, embodied AI, persistent memory architecture, multimodal binding

1. Introduction

Many contemporary AI systems depend on centralized training pipelines, large-scale labeled datasets, and cloud-based inference or coordination. While this approach has produced powerful capabilities, it is not the only possible path toward adaptive intelligence. It also leaves open important questions regarding autonomy, energy efficiency, privacy, persistence, and long-horizon adaptation on small devices.

ORÓMA explores an alternative direction: an offline-first architecture that operates continuously on edge hardware, accumulates experience over time, replays and consolidates that experience, and adapts its internal structures without requiring a conventional centralized retraining regime.

The project is motivated by three practical goals:

1. Persistent local cognition on low-power hardware.
2. Continual adaptation without dependence on cloud-scale retraining loops.
3. Structured memory formation through episodic accumulation, replay, and binding rather than one-shot batch optimization alone.

ORÓMA is therefore not positioned as a replacement for large-scale language or vision foundation models. Instead, it is positioned as an architectural exploration of persistent, edge-deployed, memory-centric cognition.

2. Problem Statement

Edge systems face a persistent tension between computational limits and adaptive behavior. Systems deployed on small hardware must often choose between being lightweight but rigid, adaptive but cloud-dependent, or locally autonomous but memory-poor.

A core problem is that many edge deployments lack a durable internal mechanism for remembering temporally structured experience, replaying and consolidating that experience, relating events across modalities and tasks, and adapting behavior over long time horizons.

ORÓMA addresses this problem by making persistent episodic memory a first-class architectural element rather than a peripheral logging feature.

3. Core Concepts

3.1 Snap

A Snap is a temporally localized observation unit. A Snap is not merely a raw frame or a single sensor sample; it is a compact observation entity that can include perceptual, contextual, temporal, and relation-bearing information.

Snapshots are intended to support local state capture, lightweight summarization, and later replay and reinterpretation.

3.2 SnapChain

A SnapChain is a temporally ordered sequence or structured linkage of Snaps representing an episode or partial episode. SnapChains are the central storage and replay units of ORÓMA.

They provide temporal continuity, event grouping, replayability, and a substrate for later consolidation, abstraction, and transfer.

3.3 Day / Dream Model

ORÓMA uses a two-phase processing model.

Day mode emphasizes live perception, immediate action, runtime policy application, and online accumulation of episodes.

Dream mode emphasizes replay, restructuring, consolidation, compression, and longer-horizon integration of past experiences.

This separation reflects the practical need to distinguish between immediate interaction loops and slower internal memory work.

3.4 Replay

Replay is a core mechanism in ORÓMA. Past episodic structures are revisited to support consolidation, pattern strengthening, policy refinement, and memory reorganization.

Replay is not treated as a minor auxiliary feature. It is part of the central learning and persistence strategy.

3.5 Binding

Binding refers to the architectural mechanisms that relate different signals, events, entities, and contexts into structured internal associations. In ORÓMA, binding is relevant across perceptual elements, time, actions, policy outcomes, and multimodal event structure.

Binding matters because persistent intelligence requires not only storing events, but also relating them in useful ways across time and context.

4. Architecture Overview

ORÓMA is organized as an edge-first architecture built around persistent memory, replay, and runtime adaptation.

4.1 Persistent Episodic Memory Layer

The memory layer stores and manages episodic experience in Snap- and SnapChain-oriented form. It supports durable local accumulation, replay selection, compression and restructuring, and later reuse.

4.2 Replay and Consolidation Layer

During Dream-like phases or replay windows, ORÓMA processes accumulated experience with the goal of consolidating signal, reducing redundancy, improving policy structures, and preserving meaningful long-horizon information.

4.3 Policy and Decision Layer

ORÓMA includes policy-oriented mechanisms that operate over structured state/action spaces and can be applied across multiple domains. The architecture supports iterative accumulation of experience rather than assuming a single static model update cycle.

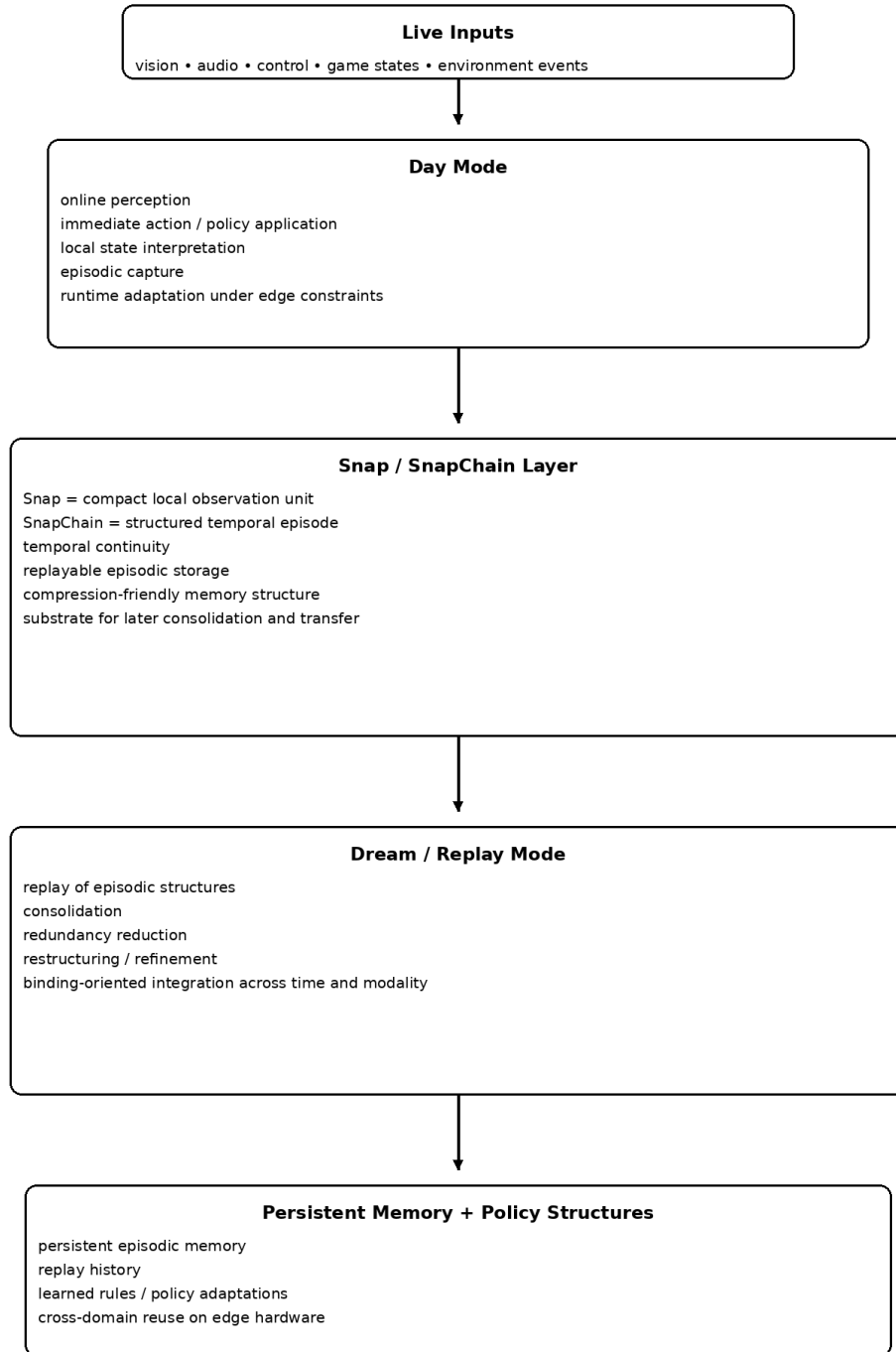
4.4 Edge Runtime and Single-Writer Stability

A practical ORÓMA design principle is runtime stability on edge hardware. This includes bounded work patterns, centralized or coordinated write paths, robust long-running operation, and explicit handling of operational constraints.

4.5 Embodiment and Real-World Coupling

ORÓMA is not purely abstract. The architecture has been exercised in domains that include embodied perception and control, such as PTZ attention loops, where memory, observation, and action interact in real time.

Figure 1 — ORÓMA Architecture Overview



High-level ORÓMA architecture. Live inputs are processed in Day mode, encoded into Snap- and SnapChain-based episodic structures, and later consolidated in Dream/Replay mode into persistent memory and adaptive policy structures.

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5. Key Contributions

This defensive publication claims the following architectural contributions as the central novelty of ORÓMA.

5.1 Offline-First Persistent Episodic Memory on Edge Hardware

ORÓMA treats persistent episodic memory as a primary architectural mechanism for a low-power edge system rather than a peripheral logging or analytics feature.

5.2 Snap / SnapChain as Operational Memory Units

The use of Snap and SnapChain structures as replayable, structured episodic units is a central design element.

5.3 Day/Dream Separation for Continual Adaptation

The explicit separation between online action-oriented processing and replay/consolidation-oriented processing is a major architectural principle.

5.4 Replay as a Core Learning Primitive

Replay is embedded as a central mechanism for consolidation and adaptation, not merely as a debugging or archival feature.

5.5 Binding-Oriented Memory Integration

Binding mechanisms are treated as an essential part of persistent cognition, supporting the integration of events, signals, and contexts across time and modality.

5.6 Edge-Centric Runtime Realism

ORÓMA is grounded in actual runtime constraints on edge hardware, including long-duration operation, bounded budgets, and stable write/control flows.

6. Memory and Consolidation Cycle

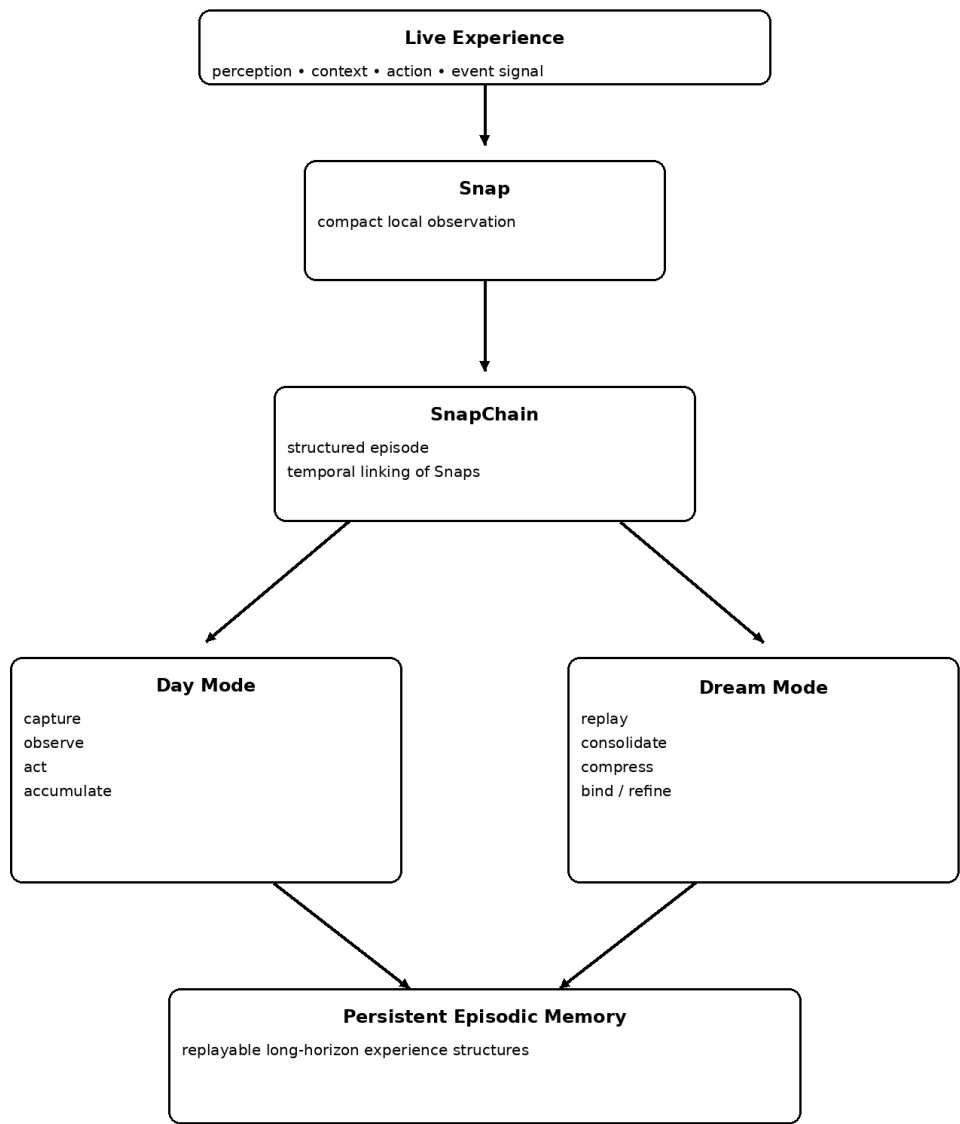
ORÓMA's central memory cycle can be summarized as follows.

Live experience is encoded into Snaps. Snaps are linked into SnapChains. These SnapChains accumulate during runtime and preserve temporal structure. During replay or Dream-like phases, the system revisits them for consolidation, compression, and structural integration.

This allows the system to preserve long-horizon experience in a form that remains operationally useful rather than relegating it to passive logging.

The importance of this cycle is that it supports persistence without requiring a cloud-based retraining loop. It makes local long-term adaptation feasible on edge hardware.

Figure 2 — ORÓMA Memory and Consolidation Cycle



Core ORÓMA memory cycle. Live experience is encoded into Snaps, linked into SnapChains, accumulated during Day mode, and revisited in Dream mode for replay, consolidation, compression, and binding.

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7. Binding and Replay as Priority Claims

Two areas are especially central to ORÓMA's novelty and therefore especially important to document as prior art.

7.1 Binding Architecture

Binding in ORÓMA is not limited to feature fusion. It is intended as a structured mechanism for persistent internal association across events, contexts, perception, action, and replay.

7.2 Universal Replay

Replay is treated as a general architectural process, not a narrow task-specific heuristic. The replay concept in ORÓMA is intended to be reusable across domains and internal subsystems.

These two themes, binding and replay, should be regarded as core pillars of the project's novelty.

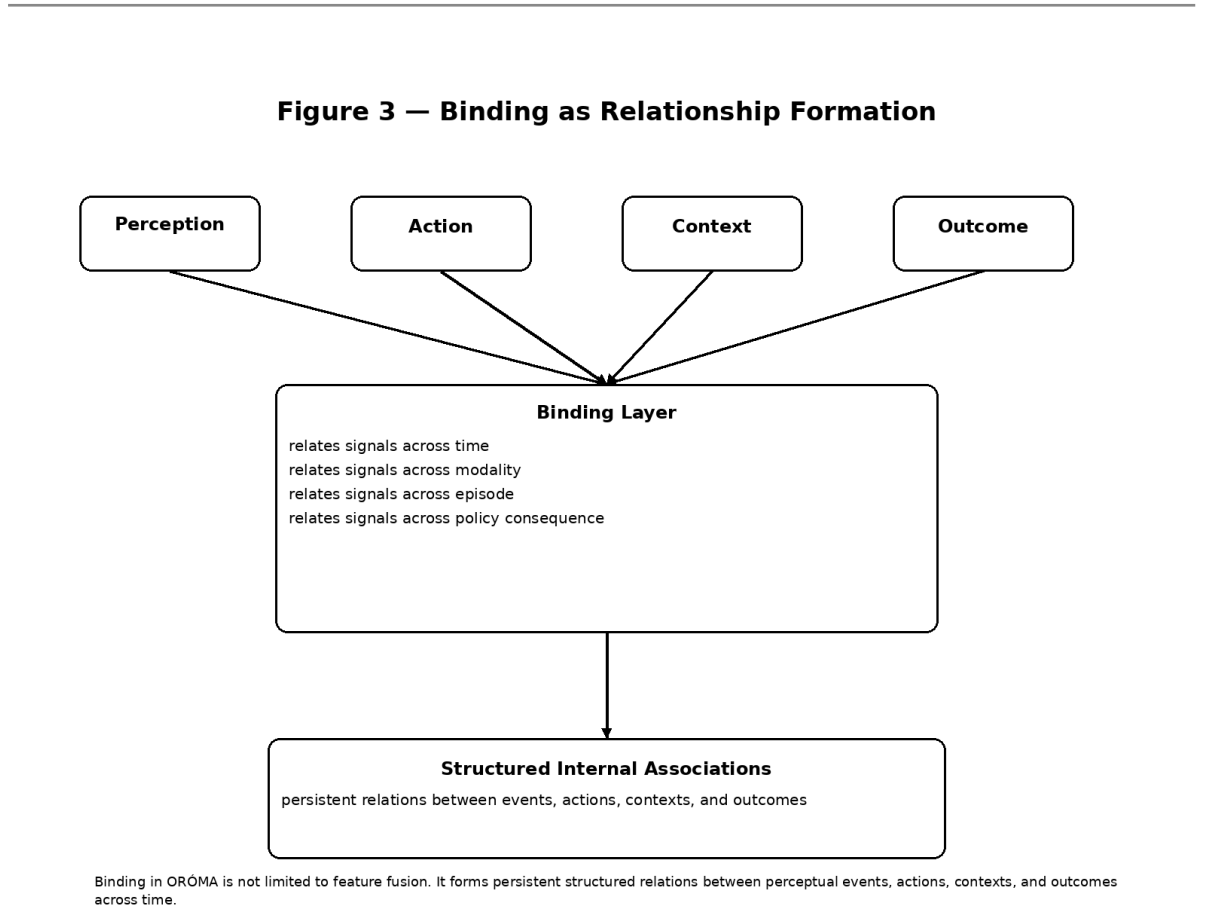


Figure 3. Binding in ORÓMA forms persistent structured relations between perceptual events, actions, contexts, and outcomes across time.

8. Implementation Status

ORÓMA is not only a conceptual sketch. The project has been implemented and iteratively extended on practical edge hardware, including Raspberry Pi-based deployment.

The documented implementation direction includes persistent episodic storage, replay and Dream-style processing, UI and observability layers, policy-oriented experimental domains, PTZ attention and video-related control, and multi-domain runtime experimentation.

The system has been exercised across multiple task domains, including game-like environments, memory and replay workflows, perception and control loops, and embodied attention behavior.

This implementation status matters because it distinguishes ORÓMA from a purely speculative architecture proposal.

9. Edge Deployment Perspective

ORÓMA is explicitly designed in the context of edge deployment rather than cloud-scale infrastructure. This leads to several practical constraints and design consequences:

- storage and write-path discipline matter,
- bounded runtime is necessary,
- observability is operationally important,
- architecture must tolerate long-lived incremental change.

This perspective informs both the implementation style and the conceptual design of the system.

10. Limitations and Deliberate Non-Disclosure

ORÓMA is an active, evolving research and engineering project. Not all internal mechanisms are fully mature, and not all implementation details are disclosed in this document.

The following points remain open research or engineering topics:

- stronger multimodal binding quality,
- further refinement of converting pressure or attention into robust task success,
- improved long-horizon internal goal formation,
- deeper generalization across heterogeneous domains.

In addition, some internal formulas, thresholds, operational heuristics, and infrastructure details are deliberately not fully disclosed here. This publication is intended to establish architectural priority and conceptual structure, not to reveal every implementation-specific optimization.

11. Conclusion

ORÓMA presents an alternative path for adaptive AI systems: not a cloud-dependent, scale-first paradigm, but an offline-first, memory-centric, replay-enabled architecture for edge cognitive agents.

Its central ideas are:

- persistent episodic memory,
- Snap and SnapChain-based experience structures,
- Day/Dream phase separation,
- replay-driven consolidation,
- binding-oriented long-horizon integration.

The project should be understood as both a practical edge architecture under active development and a conceptual contribution to persistent adaptive cognition outside the dominant cloud-scaling model.

This document is published as a defensive disclosure of the ORÓMA architecture and its central mechanisms.